

Rafay Mirza

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PROFESSIONAL SUMMARY

Full-stack developer pursuing a B.S. in Artificial Intelligence at the University of South Florida, with hands-on experience building and deploying production web applications. Skilled across the full stack — from React frontends and RESTful APIs to cloud infrastructure on AWS EC2. Experienced integrating AI/ML components, including RAG pipelines and LLM APIs, into end-to-end systems.

EDUCATION

University of South Florida Tampa, FL *B.S. Artificial Intelligence, Bellini College of AI, Cybersecurity & Computing. 2029*

SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, Git

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, RESTful API, MongoDB, Framer Motion

Infrastructure & Cloud: AWS EC2, MongoDB Atlas, Firebase Auth, Nginx, PM2, Linux/Ubuntu

Certifications: AWS Solutions Architect Associate (In Progress)

PROJECTS

AI Document Assistant (RAG-Based)

Full-Stack Developer

- Built an end-to-end RAG-based document QA system using FastAPI, FAISS as vector database, and LLaMA 3 via Groq, enabling semantic search over 1,000+ text chunks parsed from uploaded PDFs.
- Developed a React frontend with real-time chat UI and Markdown rendering, supporting 100+ user queries per session through integrated backend APIs.

Memory Wars (Multiplayer Live Web Game)

Full-Stack Developer

- Engineered real-time multiplayer gameplay using WebSocket (Socket.io) with synchronized game state, live chat, and player presence tracking across concurrent sessions.
- Built RESTful APIs in Express.js for game logic, leaderboard, and score persistence; integrated Firebase Auth with email, Google OAuth, and guest login support.
- Deployed production system on AWS EC2 with Nginx reverse proxy and PM2 process manager for 24/7 uptime; used MongoDB Atlas for the database and React with Framer Motion for a responsive UI.

EXPERIENCE

Software & Outreach Lead Jan 2025 – Jul 2025 *Truce Foundation Islamabad, PK* • Redesigned the NGO's web platform to improve accessibility for 500+ rural students gaining first-time access to digital educational resources across underserved communities.

- Collaborated with leadership on technical roadmap and drove digital outreach strategy, increasing community engagement and donor reach by 120%.

ACTIVITIES & LEADERSHIP

National Solutions Convention

Computer Science Events Officer

- Led on-site event management and invigilation; designed problem sets and test cases for computer science competitions.